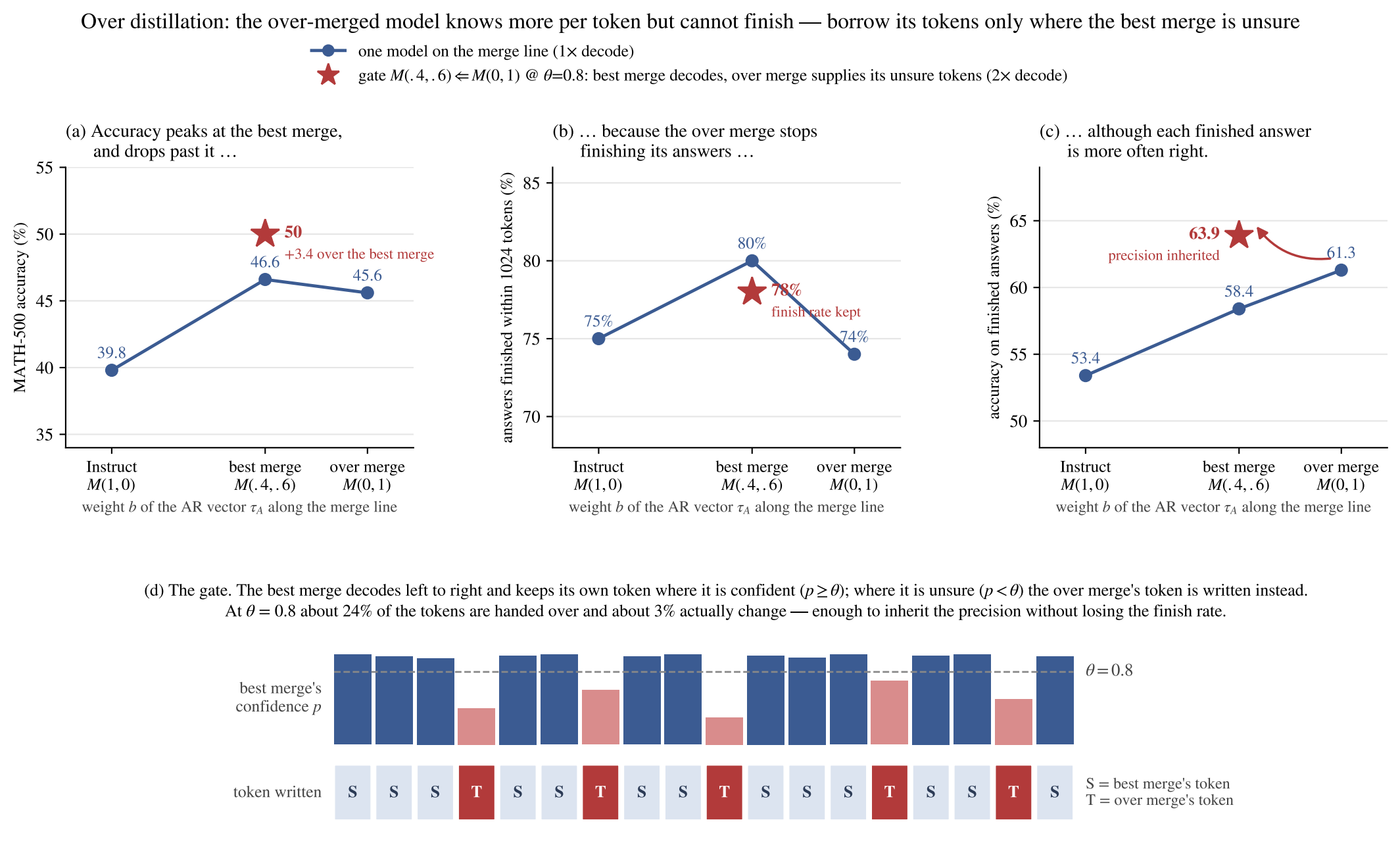
Over distillation: the over-merged model knows more per token but cannot finish — borrow its tokens only where the best merge is unsure
one model on the merge line (1× decode)
gate $M(.4,.6) \Leftarrow M(0,1)$ @ θ=0.8: best merge decodes, over merge supplies its unsure tokens (2× decode)
(a) Accuracy peaks at the best merge, and drops past it …
MATH-500 accuracy (%)
55
50
45
40
35
39.8
46.6
50
+3.4 over the best merge
45.6
Instruct $M(1,0)$
best merge $M(.4,.6)$
over merge $M(0,1)$
weight b of the AR vector τ_A along the merge line
(b) … because the over merge stops finishing its answers …
answers finished within 1024 tokens (%)
85
80
75
70
75%
80%
78%
finish rate kept
74%
Instruct $M(1,0)$
best merge $M(.4,.6)$
over merge $M(0,1)$
weight b of the AR vector τ_A along the merge line
(c) … although each finished answer is more often right.
accuracy on finished answers (%)
65
60
55
50
53.4
58.4
63.9
precision inherited
61.3
Instruct $M(1,0)$
best merge $M(.4,.6)$
over merge $M(0,1)$
weight b of the AR vector τ_A along the merge line
(d) The gate. The best merge decodes left to right and keeps its own token where it is confident ($p \geq \theta$); where it is unsure ($p < \theta$) the over merge's token is written instead.
At $\theta = 0.8$ about 24% of the tokens are handed over and about 3% actually change — enough to inherit the precision without losing the finish rate.
best merge's confidence p
$\theta = 0.8$
token written
S S S T S S T S S T S S S T S S T S
S = best merge's token
T = over merge's token